

# Action-Contract Grouping for B-rep Connector Poses: Separating Geometric Symmetry from Local Assembly Action Equivalence

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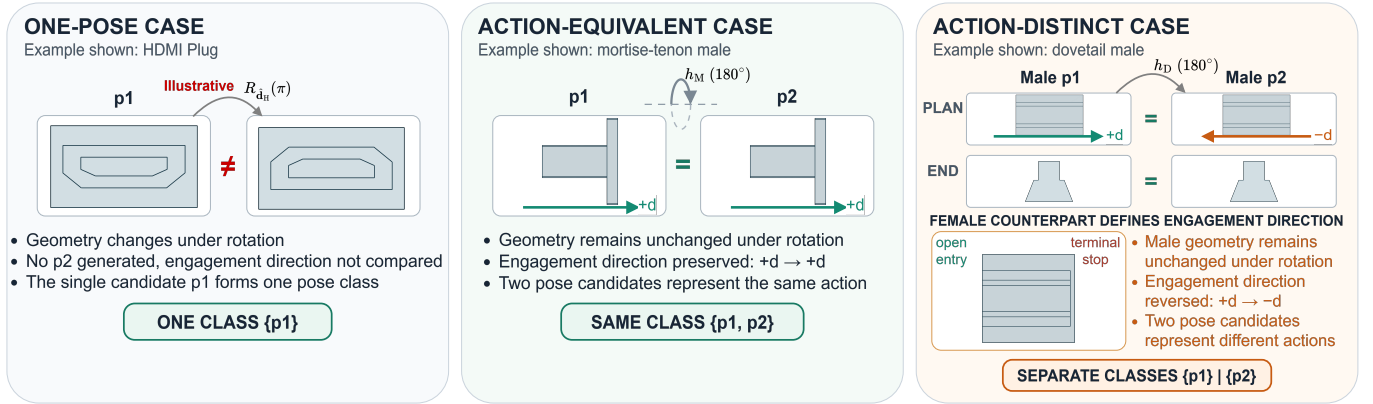


Fig. 1. The three evaluated pose-grouping cases. Curved arrows compare connector-specific rotations; they are not insertion motions. In the one-pose HDMI case (left), the illustrative rotation changes the geometry, so no second pose candidate is generated. In the mortise-tenon case (centre),  $h_M$  preserves the geometry and approved action contract, including signed engagement direction, so  $p_1$  and  $p_2$  share a class. In the dovetail-male case (right),  $h_D$  preserves the male geometry but reverses the direction defined by the female counterpart's open entry and terminal stop, so the poses remain separate. Rotation axes are connector-specific; only geometry-preserving transformations proceed to comparison of the complete approved action contract.

**Abstract**—In robotic assembly, connectors such as joints, plugs and sockets can define relationships between components. Matching a registered boundary-representation (B-rep) connector template to component geometry can, however, return multiple geometrically valid poses when the connector is symmetric. An assembly system must therefore decide whether these poses represent the same or different local actions.

This paper introduces action-contract grouping, in which pose candidates representing the same local assembly action form a pose class. Two poses share a class only when their relative transform matches an approved geometric self-symmetry that preserves the connector's local action contract, including directional constraints. The method returns one robot-agnostic record per pose class. This represents the same local action once whilst keeping every candidate pose available.

Evaluation used 18 primary scenes covering single-pose, action-equivalent and action-distinct cases. Compared with withheld references derived from connector specifications, the method organised 27 poses into 21 required classes and reproduced all 18 groupings. Therefore, within this registered-CAD setting, the method produced six fewer pose-class records than the one-record-per-pose baseline, whilst preserving every directionally distinct alternative. In three cases, geometry-only grouping incorrectly combined poses representing different local assembly actions into the same class, whilst the proposed method correctly assigned them to separate classes. Keeping these actions separate preserves distinct assembly alternatives for later planning decisions.

**Index Terms**—Computer-aided design, geometric modeling, pose grouping, robotic assembly

## I. INTRODUCTION

Geometric symmetry creates a representation problem in robotic assembly once candidate poses have been generated [1]. Matching a symmetric geometric template can return several geometrically valid poses for the same connector instance [2], [3]. A system must then decide whether these poses represent the same local assembly action or alternatives that must remain separate. Geometrically equivalent poses can specify different engagement directions. Grouping these poses as one action may therefore hide an assembly alternative from later planning stages. The focus of this paper is therefore the post-localisation grouping decision, with the preceding connector-recognition and pose-generation stages held fixed.

In this paper, a registered connector is a versioned canonical STEP/B-rep template paired with an approved role-specific local action contract, with the term connector being used broadly to include mechanical joints as well as electrical plugs and sockets.

Selecting one pose may, however, discard a directionally distinct alternative. Emitting one record per pose retains every alternative but can duplicate an action already represented. Geometry-only grouping can remove this duplication but may also combine poses when the relative symmetry reverses the signed engagement direction. Fig. 1 illustrates these cases.

This paper introduces action-contract grouping and makes three contributions:

- 1) A grouping rule for deciding which symmetry-related poses represent the same local assembly action. This checks whether the corresponding geometric self-symmetry preserves the connector's approved local action contract.
- 2) A robot-agnostic representation with one record per pose class. Each class represents one local action whilst retaining every member pose.
- 3) A controlled comparison with alternative grouping strategies using the same candidate poses and withheld reference groupings derived from connector specifications.

The pose classes describe local action equivalence and do not establish global assembly feasibility.

Evaluation comprises 18 primary scenes covering cases with one pose, two action-equivalent poses, or two action-distinct poses. Separate topology-changing scenes and repeated-connector controls examine the exact-matching stage. The pose-class records provide a robot-agnostic intermediate representation to support future integration with planning, skill-assignment and execution systems.

## II. RELATED WORK

BRepNet segments faces directly on B-rep models [4], and Wang et al. identify assembly-interface faces and classify their types [5]. AutoMate predicts CAD mates from parametric B-reps [6]. JoinABLE predicts compatible B-rep entities and joint parameters for bottom-up assembly [7]. These methods operate at the face, mate, or joint level, but do not partition multiple exact poses of an already-localised registered connector.

Brégier et al. introduce symmetry-aware measures for evaluating 3D pose estimates [2], while EPOS estimates poses of objects with global or partial symmetries [3]. For assembly, Liu and Popplestone use symmetry groups to analyse assembly contact states and motions [1]. Mates2Motion infers motion degrees of freedom and axes from CAD mates [8], TAX-Pose learns task-specific cross-pose relationships [9], and 2BY2 estimates pairwise assembly poses while accounting for symmetry and functional or spatial relations [10]. The problem considered here begins after localisation: the input is a fixed set of exact registered-CAD pose candidates, and the task is to determine which candidates represent the same local action under the approved local action contract.

At the system level, Pane et al. use authored CAD constraints to select and parameterise sensor-based assembly skills [11]. Manual2Skill++ extracts connector relations from instruction manuals within a simulated pipeline from task understanding to execution [12]. The present work is intended to contribute one post-localisation component of a broader CAD-centred flexible-assembly architecture: deciding whether symmetry-related connector poses represent the same local action and should be grouped. The resulting pose classes provide a robot-agnostic representation for later assembly-system stages.

## III. ACTION-CONTRACT GROUPING

For connector instance  $i$ , let  $\mathcal{P}_i$  denote the finite set of pose candidates.

Using column vectors,  ${}^A T_B = (R_{AB}, t_{AB}) \in \text{SE}(3)$ , with  $R_{AB} \in \text{SO}(3)$  and  $t_{AB} \in \mathbb{R}^3$ , maps coordinates from  $B$  to  $A$  as  ${}^A x = R_{AB} {}^B x + t_{AB}$ . Each candidate  $p \in \mathcal{P}_i$  is represented by  $T_p \equiv {}^S T_C^{(p)}$ , which maps the canonical connector frame  $C$  into the scene frame  $S$ . The connector registry defines a finite set of proper geometric self-symmetries  $H_i^{\text{geom}} \leq \text{SE}(3)$  using standard  $\text{SO}(3)$  and  $\text{SE}(3)$  matrix conventions [13]. For each stored symmetry  $h = (R_h, t_h)$ , the rotation matrix is numerically validated by

$$\max_{j,k} |(R_h^T R_h - I)_{jk}| \leq 10^{-9} \quad \text{and} \quad |\det R_h - 1| \leq 10^{-9}. \quad (1)$$

The set is also checked for identity, closure and inverses using the fixed transform-matching tolerances. Because a symmetry is expressed in the canonical connector frame, it acts on a candidate transform by right multiplication  $T \mapsto Th$ . Therefore, for two candidates  $p, q \in \mathcal{P}_i$ ,

$$h_{pq} = T_p^{-1} T_q. \quad (2)$$

The approved local action contract and its validation record are defined as

$$C_i^{\text{act}} = (F_i, A_i, O_i, M_i, r_i), \quad \mathcal{G}_i = (S_i, P_i, v_i). \quad (3)$$

Here,  $F_i$  denotes the local action frame, whose pose in the canonical connector frame  $C$  is given by  ${}^C T_{F_i} \in \text{SE}(3)$ . The terms  $A_i$ ,  $O_i$  and  $M_i$  specify the anchor, orientation observables, and ordered `axis3/twist6` phases respectively, whilst  $r_i$  records the interface identity and phase-role polarity. Interface role is fixed by the template-to-contract binding, whereas phase-role polarity is compared exactly. The validation record  $\mathcal{G}_i$  contains the approved support reference and hash, review/provenance state and version information. Provided that the required records are valid, preservation under  $h$  is defined by

$$\text{Preserves}(h, C_i^{\text{act}}) := \text{Compare}(h \cdot C_i^{\text{act}}, C_i^{\text{act}}). \quad (4)$$

After re-expression in  $C$ , only the geometric descriptors are transformed by  $h$ . Compare checks the transformed contract against the original using the preservation criteria in Table I. Geometric descriptors are compared using their declared freedoms and numerical tolerances. Non-geometric fields, including descriptor types, directionality declarations and phase-role polarity, must match. Ordered phases require a one-to-one correspondence that satisfies all phase checks while preserving their declared ordering. For a valid contract, Compare returns true only when every applicable check passes, and false otherwise.

If required information is missing or invalid, preservation is undefined and pose-class output is blocked. This is not treated as evidence that the poses are action-distinct.

For example, the mortise-tenon symmetry  $h_M$  in Fig. 1 preserves the signed engagement direction ( $+d \mapsto +d$ ) and

TABLE I  
FIXED PRESERVATION CHECKS.

| Item              | Preserved when                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orientation       | Angular tolerance passes. Directed/oriented signs are retained; undirected/unoriented signs may reverse. Type, polarity and required status match exactly.                                                                                                                                                                                                                                                                          |
| Point anchor      | Point displacement is within the anchor tolerance; no coordinate freedom is permitted.                                                                                                                                                                                                                                                                                                                                              |
| Line/plane anchor | Residual (6) and angular test pass; displacement is only along the line or within the plane.                                                                                                                                                                                                                                                                                                                                        |
| Ordered phases    | Exhaustive search finds a bijection satisfying:<br><b>Order:</b> phase type, order and predecessors match.<br><b>Motion:</b> representation, directionality and role polarity match. Signs may reverse only when declared undirected/bidirectional.<br><b>Bounds:</b> units and bound semantics match; finite bounds agree within the stated tolerances.<br><b>Support:</b> required support is unchanged in kind, status and hash. |
| Validation        | Support reference/hash, review status, provenance and version are valid. Missing required support blocks output. These records are not transformed.                                                                                                                                                                                                                                                                                 |

passes all other contract checks. Compare therefore returns true, confirming that this symmetry preserves the local action contract. The dovetail-male symmetry  $h_D$  reverses the signed engagement direction ( $+d \mapsto -d$ ). This fails the signed-direction check, so Compare returns false and the two poses remain in separate classes.

Writing  ${}^C T_{F_i} = (R_{CF_i}, t_{CF_i})$  for the transform from the local action frame  $F_i$  to the canonical connector frame  $C$ ,  $R_{CF_i}$  rotates vectors into  $C$  and  $t_{CF_i}$  gives the origin of  $F_i$  expressed in  $C$ . Points, directions and angular-first twist coordinates authored in  $F_i$  are then re-expressed in  $C$  as

$${}^C p = R_{CF_i} {}^{F_i} p + t_{CF_i}, \quad {}^C d = R_{CF_i} {}^{F_i} d, \quad {}^C \xi = \text{Ad}_{{}^C T_{F_i}} {}^{F_i} \xi. \quad (5)$$

For  $\xi = [\omega^\top, v^\top]^\top \in \mathbb{R}^6$ , where  $\omega, v \in \mathbb{R}^3$ , the associated Lie-algebra matrix is

$$\widehat{\xi} = \xi^\wedge = \begin{bmatrix} [\omega]_\times & v \\ 0^\top & 0 \end{bmatrix} \in \mathfrak{se}(3).$$

For  $T = (R, t) \in \text{SE}(3)$ , the coordinate adjoint is defined by

$$(\text{Ad}_T \xi)^\wedge = T \xi^\wedge T^{-1}, \quad \text{Ad}_T = \begin{bmatrix} R & 0 \\ [t]_\times R & R \end{bmatrix}.$$

Under the angular-first convention, the transformed twist components satisfy  $\omega' = R\omega$  and  $v' = Rv + [t]_\times R\omega$ . The translation between frame origins therefore contributes to the transformed linear component of the twist. Solà *et al.* use the linear-first ordering  $[\rho^\top, \theta^\top]^\top$ ; the displayed adjoint is its block-reordered angular-first form [13]. Writing the re-expressed quantities with a  $C$  subscript, a symmetry  $h = (R_h, t_h) \in H_i^{\text{geom}}$  gives  $p'_C = R_h p_C + t_h$ ,  $d'_C = R_h d_C$ ,  $n'_C = R_h n_C$ , and  $\xi'_C = \text{Ad}_h \xi_C$ . Therefore, an `axis3` descriptor is direction-only

and uses the rotational action, whereas a `twist6` descriptor uses the adjoint.

An unconstrained anchor has no positional constraint, whereas a point anchor uses the full rigid-body transformation, and line and plane anchors permit displacement only along the line and within the plane, respectively. For a line  $(p_C, d_C)$  and plane  $(p_C, n_C)$ , with unit  $d_C$  and  $n_C$ , the perpendicular residuals are

$$\delta_{\text{line}} = \|(I - d_C d_C^\top)(p'_C - p_C)\|_2, \quad \delta_{\text{plane}} = |n_C^\top (p'_C - p_C)|. \quad (6)$$

For either anchor type, the transformed direction or normal is compared with its original value using the directionality declared by the contract. The remaining preservation criteria are summarised in Table I.

A relative transform is matched to  $H_i^{\text{geom}}$  using translation and rotation tolerances of  $10^{-5}$  mm and  $10^{-4}$  degrees respectively, and is accepted only if exactly one symmetry matches. Action comparison uses an anchor-distance tolerance of  $10^{-4}$  mm, and an angular tolerance of  $10^{-3}$  degrees for orientation vectors and plane normals. Two scalar bounds  $a$  and  $b$ , expressed in the unit  $u$ , are considered equal when

$$|a - b| \leq \max(\epsilon_{\text{abs},u}, 10^{-9} \max(|a|, |b|)),$$

where  $\epsilon_{\text{abs},u}$  is  $10^{-6}$  mm,  $10^{-8}$  rad or  $10^{-9}$  for length, angular or dimensionless quantities, respectively. These thresholds were fixed before evaluation.

For each unordered pair of distinct candidates, let  $(a, b)$  denote its deterministic ordering, set  $h_{ab} = T_a^{-1} T_b$ , and let  $\pi_i(h_{ab})$  denote the unique element of  $H_i^{\text{geom}}$  matching  $h_{ab}$  within the stated tolerances. If no unique match exists,  $\pi_i(h_{ab})$  is undefined. Set  $p \sim_i p$  for every  $p \in \mathcal{P}_i$ . For  $p \neq q$ , the implementation defines

$$p \sim_i q \iff \pi_i(h_{ab}) \text{ is defined } \wedge \text{Preserves}(\pi_i(h_{ab}), C_i^{\text{act}}). \quad (7)$$

This decision is assigned to  $(p, q)$  and  $(q, p)$ , so the relation is symmetric by construction, and the resulting finite relation is then checked for transitivity. If the check fails, no pose classes are emitted; otherwise  $Q_i = \mathcal{P}_i / \sim_i$  is the set of pose classes. When the output is valid, each candidate belongs to exactly one class, and no group structure is implied. Each pose-class record contains a deterministic representative, all member-pose identifiers, the action frame expressed in scene coordinates, the interface role, and the approved contract hash, while the complete pose-candidate data remain available.

Define the action-preserving symmetries as

$$H_i^{\text{act}} = \{h \in H_i^{\text{geom}} : \text{Preserves}(h, C_i^{\text{act}})\}.$$

In the idealised exact-transform case, where, for each distinct candidate pair,  $\pi_i(h_{ab})$  is defined if and only if  $h_{ab} \in H_i^{\text{geom}}$  and then returns  $h_{ab}$ , (7) is an equivalence relation whenever  $H_i^{\text{act}}$  is a subgroup of  $H_i^{\text{geom}}$ . If the candidate-to-transform map  $p \mapsto T_p$  is additionally a bijection from  $\mathcal{P}_i$  onto  $T_0 H_i^{\text{geom}}$ , each pose class corresponds to a right  $H_i^{\text{act}}$ -orbit. Complete

coverage is required only for this orbit interpretation, not for classification of the finite candidate set. The implementation validates  $H_i^{\text{geom}}$  and checks the resulting finite relation for transitivity, but does not separately certify  $H_i^{\text{act}}$  as a subgroup for each contract.

#### IV. CONTROLLED EVALUATION

The exact-matching stage was fixed before evaluation and shared by all methods. Six registered connector roles were evaluated, each with a predefined local action contract: HDMI plug and socket, dovetail male and female, and mortise–tenon male and female. HDMI plug, HDMI socket and dovetail female form the one-pose cases; the mortise–tenon roles form the action-equivalent cases; and dovetail male forms the action-distinct case.

Table II compares the grouping strategies. The single-pose baseline (B0) retains the first deterministically ordered pose, R0 assigns every pose to a separate class, G0 returns one class per geometric orbit, and P2 applies (7). The control A1 repeats P2 while treating signed phase directions as unsigned. For A2, the required compatible-counterpart support is removed from the applicable dovetail-male contract to test whether pose-class output is blocked.

The suite comprises 27 multi-body CAD scenes: 18 primary scenes, six topology-changing negative scenes, and three repeated-connector controls containing unrelated distractor geometry. The 18 primary connector instances yield 27 candidate poses and 21 reference classes, including nine one-pose, six action-equivalent, and three action-distinct cases. Including the repeated controls, the conforming scenes contain 24 connector instances. Notably, the negative scenes contain no expected conforming connector instances, while the repeated controls assess the exact-matching stage and are excluded from the primary pose-grouping comparison.

For each primary scene, an author-reviewed reference partition was prepared independently of P2 from connector specifications, assigning reference poses to the required classes. These partitions were withheld from P2; neither they nor the resulting scores were used to tune the comparator or tolerances, which were fixed before scene generation. Host geometry, connector placement and orientation, body ordering, and repeated-connector and distractor composition vary, while the six templates and contracts remain fixed. A single comparator with no connector-name-specific branch was then applied to all six contracts.

The A1 CAD control isolates the effect of signed direction. The other preservation criteria in Table I are exercised by 13 analytic witness cases and their associated software tests, and are not evaluated through additional connector-family scenes.

Within the fixed transform tolerances, the scorer maps each uniquely matched predicted pose one-to-one to a reference pose. A reference class contributes to required-class coverage when it contains the mapped representative of at least one predicted class. Exact agreement instead compares the complete mapped pose-class partition with the reference partition.

TABLE II  
PRIMARY GROUPING RESULTS. RC: REQUIRED CLASSES REPRESENTED; FM: FALSE MERGES; USP: UNREPRESENTED SAME-CLASS PAIRS; EC: EXCESS CLASSES.

| Method                      | RC           | FM       | USP      | EC       | Exact        |
|-----------------------------|--------------|----------|----------|----------|--------------|
| Single pose (B0)            | 18/21        | 0        | 6        | 0        | 9/18         |
| All poses (R0)              | 21/21        | 0        | 6        | 6        | 12/18        |
| Geometry only (G0)          | 18/21        | 3        | 0        | 0        | 15/18        |
| <b>Action contract (P2)</b> | <b>21/21</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>18/18</b> |
| No direction sign (A1)      | 18/21        | 3        | 0        | 0        | 15/18        |

#### V. RESULTS AND LIMITATIONS

Table II shows the characteristic failure mode of each grouping strategy. The exact-matching stage returned 27 candidate poses for 21 primary reference classes. B0 retained 18 poses, omitting one directional alternative in each of the three dovetail-male cases, and one member of each of the six action-equivalent mortise–tenon pairs. R0 retained all 27 poses but split each mortise–tenon pair into separate classes, producing six unrepresented same-class pairs and six excess classes. G0 removed these duplicate classes but falsely merged the two poses in each of the three action-distinct dovetail-male cases. P2 returned 21 classes and reproduced all 18 reference groupings exactly, with no false merges, unrepresented same-class pairs, or excess classes. A1 reproduced the same three false merges, showing that signed direction was the distinguishing contract field in the CAD cases designed to isolate it.

The shared exact-matching stage recovered all 18 primary connector instances, rejected all six negative scenes, and recovered all six repeated-control connector instances without distractor matches. In A2, removing the required compatible-counterpart support caused all five applicable dovetail-male instances to return no pose-class output. These comprised three primary instances and two repeated-control instances. The control did not apply to the remaining 19 conforming instances. These five cases represent repeated evaluations of one role-specific local action contract rather than five independent contracts.

The evaluation is closed-world. Connector templates, local action contracts, frames, motion semantics, and support records are predefined rather than inferred, and exact matching assumes unchanged nominal B-rep topology and geometry. The study therefore does not evaluate manufacturing variation, tolerance-aware matching, fused-body recognition, continuous symmetry, or generalisation to unseen connector families and contract types. Local assembly-action equivalence also does not establish global access, grasp, collision, reachability, planning, or execution feasibility.

Within the evaluated registered-CAD setting, action-contract grouping reproduced all 21 required primary classes and all 18 reference partitions without false merges, unrepresented same-class pairs or excess classes. The resulting robot-agnostic pose-class records provide a basis for future integration with planning, skill-assignment and execution systems.

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